26.3: Shared characters are used to construct phylogenetic trees

As we've discussed, a key step in reconstructing phylogenies is to distinguish homologous features from analogous ones (since only homology reflects evolutionary history). Next, we'll describe cladistics, a widely used set of methods for inferring phylogeny.

Cladistics

In the approach to systematics called **cladistics**, common ancestry is the primary criterion used to classify organisms. Using this methodology, biologists attempt to place species into groups called **clades**, each of which includes an ancestral species and all of its descendants. Clades, like the categories of the Linnaean system, are nested within larger clades. In Figure 26.4, for example, the cat group (Felidae) represents a clade within a larger clade (Carnivora) that also includes the dog group (Canidae).

A taxon is equivalent to a clade only if it is **monophyletic** (from the Greek, meaning "single tribe"), signifying that it consists of an ancestral species and all of its descendants **(Figure 26.10a)**. Contrast this with a **paraphyletic** ("beside the tribe") group, which consists of an ancestral species and some, but not all, of its descendants **(Figure 26.10b)**, or a **polyphyletic** ("many tribes") group, which includes distantly related species but does not include their most recent common ancestor **(Figure 26.10c)**.

Figure 26.10

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Monophyletic, paraphyletic, and polyphyletic groups.
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A phylogenetic tree illustrates three different groups.

The tree is as follows. Root is leads to nodes 1. Node 1 is labeled common ancestor 3. It splits to nodes 2 and 3. Node 2 is labeled common ancestor 1. It splits to branch tip A and node 4. Node 4 splits to branch tips B and C. Node 3 is labeled common ancestor 2. It splits to nodes 5 and 6. Node 5 splits to branch tips D and E. Node 6 splits to branch tips F and G. Common ancestor 1 and branches A, B and C are highlighted, representing group 1. Group 1, consisting of three species (A, B, C) and their common ancestor 1, is a monophyletic group (clade), meaning that it consists of an ancestral species and all of its descendants. Common ancestor 2 and branches D, E, and F are highlighted, representing group 2. Group 2 is paraphyletic, meaning that it consists of an ancestral species 2 and some of its descendants (species D, E, F) but not all of them (does not include species G). Branches A, B, C, and D are highlighted, representing group

3. Group 3, consisting of four species (A, B, C, D), is polyphyletic, meaning that the most recent common ancestor, 3, of its members is not part of the group.

Note that in a paraphyletic group, the most recent common ancestor of all members of the group *is* part of the group, whereas in a polyphyletic group the most recent common ancestor *is not* part of the group. For example, a group consisting of even-toed ungulates (hippopotamuses, deer, and their relatives) and their common ancestor is paraphyletic because it does not include cetaceans (whales, dolphins, and porpoises), which descended from that ancestor **(Figure 26.11)**. In contrast, a group consisting of seals and cetaceans (based on their similar body forms) is polyphyletic because it does not include the common ancestor of seals and cetaceans. Biologists avoid defining such polyphyletic groups; if new evidence indicates that an existing group is polyphyletic, its members are reclassified.

Figure 26.11

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Paraphyletic versus polyphyletic groups.
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A phylogenetic tree showing paraphyletic and polyphyletic groups.

The tree is as follows. Root leads to node 1. Node 1 splits to nodes 2 and 3. Node 2 represents common ancestor of even-toed ungulates. It splits to branch tip, other even-toed ungulates, and node 4. Node 4 splits to branch tips hippopotamuses and cetaceans. Node 3 splits to node 5 and branch tip, other carnivores. Node 5 splits to branch tips seals and bears. The paraphyletic group example consists of the other even-toed ungulates and the hippopotamuses, because this grouping does not include all the descendants of the common ancestor (it excludes cetaceans). The polyphyletic group example consists of the cetaceans and seals, because this group does not include the most recent common ancestor of its members.

DRAW IT *Circle the branch point that represents the most recent common ancestor of cetaceans and seals. Explain why that ancestor would not be part of a cetacean–seal group defined by similar body form.*

*For suggested answer, see*Appendix A*.*

Shared Ancestral and Shared Derived Characters

As a result of descent with modification, organisms have characters they share with their ancestors, and they also have characters that differ from those of their ancestors. For example, all mammals have backbones, but a backbone does not distinguish mammals from other vertebrates because *all* vertebrates have backbones. The backbone predates the branching of mammals from other vertebrates. Thus for mammals, the backbone is a **shared ancestral character**, a character that originated in an ancestor of the taxon. In contrast, hair

is a character shared by all mammals but *not* found in their ancestors. Thus, in mammals, hair is considered a **shared derived character**, an evolutionary novelty unique to a clade.

Note that a shared derived character can refer to the loss of a feature, such as the loss of limbs in snakes or whales. Also, keep in mind that it is a relative matter whether a character is considered ancestral or derived. A backbone can also qualify as a shared derived character, but only at a deeper branch point that distinguishes all vertebrates from other animals.

Inferring Phylogenies Using Derived Characters

Shared derived characters are unique to particular clades. Because all features of organisms arose at some point in the history of life, it should be possible to determine the clade in which each shared derived character first appeared and to use that information to infer evolutionary relationships.

To give an example of this approach, consider the set of characters shown in **Figure 26.12a** for each of five vertebrates—a leopard, turtle, frog, bass, and lamprey (a jawless aquatic vertebrate). As a basis of comparison, we need to select an outgroup. An **outgroup** is a species or group of species from an evolutionary lineage that is closely related to but not part of the group of species that we are studying (the **ingroup**). A suitable outgroup can be determined based on evidence from morphology, paleontology, embryonic development, and gene sequences. An appropriate outgroup for our example is the lancelet, a small animal that lives in mudflats and (like vertebrates) is a member of the more inclusive group called the chordates. Unlike the vertebrates, however, the lancelet does not have a backbone.

Figure 26.12

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Using derived characters to infer phylogeny.
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Using derived characters to infer phylogeny.

Part a. Character table. A 0 indicates that the character is absent; a 1 indicates that the character is present.

Part b. Phylogenetic tree for the evolution of vertebrates. The character table in a leads us to infer the phylogenetic relationships shown in the tree, which is as follows. Root leads to node 1. Node 1 splits to branch tip lancelet, outgroup, and node 2 with characteristic vertebral column. Node 2 splits to branch tip lamprey and node 3 with characteristic hinged jaws. Node 3 splits to branch tip bass and node 4 with characteristic four limbs. Node 4 splits to branch tip frog and node 5 with characteristic amnion. Node 5 splits to branch tip turtle and leopard, with characteristic hair.

Characters	Lancelet (outgroup)	Lamprey	Bass	Frog	Turtle	Leopard
Vertebral column (backbone)	0	1	1	1	1	1
Hinged jaws	0	0	1	1	1	1
Four walking legs	0	0	0	1	1	1
Amnion	0	0	0	0	1	1
Hair	0	0	0	0	0	1

The derived characters used here include the amnion, a membrane that encloses the embryo inside a fluid-filled sac (see Figure 34.26). Note that a different set of characters could lead us to infer a different phylogenetic tree.

DRAW IT*In (b), circle the most inclusive clade for which a hinged jaw is a shared* ancestral *character*.

*For suggested answer, see*Appendix A.*

Figure 26.12 Walkthrough

Using derived characters to infer phylogeny

Watch an explanation of this figure in the Figure 26.12 Walkthrough: Using derived characters to infer phylogeny.

image

Play

image

Play

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In our analysis, a character found in both the outgroup and the ingroup is assumed to be ancestral. We'll also assume that each derived character in Figure 26.12a arose only once in the ingroup. Thus, for a character that only occurs in a subset of the ingroup, we'll assume that the character arose in the lineage leading to those members of the ingroup.

By comparing members of the ingroup with each other and with the outgroup, we can determine which characters were derived at the various branch points of vertebrate evolution. In our example, *all* of the

vertebrates in the ingroup have backbones: This character was present in the ancestral vertebrate, but not in the outgroup. Now note that hinged jaws are absent in the outgroup and in lampreys, but present in all other members of the ingroup. This indicates that hinged jaws arose in a lineage leading to all members of the ingroup *except* lampreys. Hence, we can conclude that lampreys are the sister taxon to the other vertebrates in the ingroup. Proceeding in this way, we can translate the data in our table of characters into a phylogenetic tree that places all the ingroup taxa into a hierarchy based on their derived characters **(Figure 26.12b)**.

Phylogenetic Trees with Proportional Branch Lengths

In the phylogenetic trees we have presented so far, the lengths of the tree's branches do not indicate the degree of evolutionary change in each lineage. Furthermore, the chronology represented by the branching pattern of the tree is relative (earlier versus later) rather than absolute (how many millions of years ago). But in some tree diagrams, branch lengths are proportional to the amount of evolutionary change or to the times at which particular events occurred.

In **Figure 26.13**, for example, each branch length of the phylogenetic tree reflects the number of changes that have taken place in a particular DNA sequence in that lineage. Note that the total length of the horizontal lines from the base of the tree to the mouse is less than that of the line leading to the fruit fly, *Drosophila*. This indicates that fewer genetic changes have occurred in the mouse lineage than in the *Drosophila* lineage. In addition, because the mouse and fly lineages have evolved for an equal amount of time after they diverged from a common ancestor, the *rate* of change was lower in mouse lineage than in the *Drosophila* lineage.

Figure 26.13

Branch lengths can represent genetic change.

Phylogenetic tree where branch length represents degree of genetic change.

This tree was based on sequences of homologs of a gene that plays a role in development; Drosophila was used as an outgroup. Branch lengths are proportional to the amount of genetic change in each lineage; varying branch lengths indicate that the gene has evolved at different rates in different lineages.

The tree is as follows, from top to bottom. Note, branches decrease in length from top to bottom. Root leads to node 1. Node 1 splits to branch top drosophila and node 2. Node 2 splits to branch tip lancelet and node 3. Node 3 splits to branch tip zebrafish and node 4. Node 4 splits to branch tip frog and node 5. Node 5 splits to branch tip chicken and node 6. Node 6 splits to branch tips human and mouse, each the same length.

INTERPRET THE DATA *In which vertebrate lineage has the studied gene evolved most rapidly? Explain.*

In general, although the branches of a phylogenetic tree may have different lengths, among organisms alive today, all the different lineages that descend from a common ancestor have survived for the same number of years. To take an extreme example, humans and bacteria had a common ancestor that lived over 3 billion years ago. Fossils and genetic evidence indicate that this ancestor was a single-celled prokaryote. Even though bacteria have apparently changed little in their morphology since that common ancestor, there have nonetheless been 3 billion years of evolution in the bacterial lineage, just as there have been 3 billion years of evolution in the lineage that ultimately gave rise to humans.

These equal spans of chronological time can be represented in a phylogenetic tree whose branch lengths are proportional to time **(Figure 26.14)**. Such a tree draws on fossil data to place branch points in the context of geologic time. Additionally, it is possible to combine these two types of trees by labeling branch points with information about rates of genetic change or dates of divergence.

Figure 26.14

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Branch lengths can indicate time.
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Evolutionary tree with branch lengths mapped to dates based on fossil evidence. Time scale is millions of years ago, from over 500 million to present.

The previous tree is displayed with branch lengths measured by a timeline as follows.

Drosophila, 620 million years.

Lancelet, 550 million years.

Zebrafish, 410 million years.

Frog, 350 million years.

Chicken, 310 million years.

Human, 60 million years.

Mouse, 60 million years.

All values estimated.

This tree is based on the same DNA data as that in Figure 26.13, but here the branch points are dated based on fossil evidence. Thus, the branch lengths are proportional to time. Each lineage has the same total length

from the base of the tree to the branch tip, indicating that all the lineages have diverged from the common ancestor for equal amounts of time.

Maximum Parsimony and Maximum Likelihood

As the database of DNA sequences that enables us to study more species grows, the difficulty of building the phylogenetic tree that best describes their evolutionary history also grows. What if you are analyzing data for 50 species? There are 3×1076 different ways to arrange 50 species into a tree! And which tree in this huge forest reflects the true phylogeny? Systematists can never be sure of finding the most accurate tree in such a large data set, but they can narrow the possibilities by applying the principles of maximum parsimony and maximum likelihood.

According to the principle of **maximum parsimony**, we should first investigate the simplest explanation that is consistent with the facts. (The parsimony principle is also called “Occam’s razor” after William of Occam, a 14th-century English philosopher who advocated solving problems by “shaving away” unnecessary complications.) In the case of trees based on morphology, the most parsimonious tree requires the fewest evolutionary events, as measured by the origin of shared derived morphological characters. For phylogenies based on DNA, the most parsimonious tree requires the fewest base changes.

A **maximum likelihood** approach identifies the tree most likely to have produced a given set of DNA data based on certain probability rules about how DNA sequences change over time. For example, the underlying probability rules could be based on the assumption that all nucleotide substitutions are equally likely. If evidence suggests that this assumption is not correct, more complex rules could be devised to account for different rates of change among different nucleotides or at different positions in a gene.

Scientists have developed many computer programs to search for trees that are parsimonious and likely. When a large amount of accurate data is available, the methods used in these programs usually yield similar trees. As an example of one method, **Figure 26.15** walks you through the process of identifying the most parsimonious molecular tree for a three-species problem. Computer programs use the principle of parsimony to estimate phylogenies in a similar way: They examine large numbers of possible trees and identify those that require the fewest evolutionary changes.

Figure 26.15

Research Method: Applying Parsimony to a Problem in Molecular Systematics

Application

In considering possible phylogenies for a group of species, systematists compare molecular data for the species. An efficient way to begin is by identifying the most parsimonious hypothesis—the one that requires the fewest evolutionary events (molecular changes) to have occurred.

Technique

Follow the numbered steps as we apply the principle of parsimony to a hypothetical phylogenetic problem involving three closely related beetle species.

Applying parsimony to a problem in molecular systematics.

Part 1. Three species of beetles drawn, each with different colors and markings. Part 2. Three possible phylogenetic hypotheses are drawn as simple tree diagrams as follows.

- Hypothesis 1. Root leads to node 1. Node 1 splits to node 2 and branch tip 3. Node 2 splits to branch tips 1 and 2.
- Hypothesis 2. Root leads to node 1. Node 1 splits to node 2 and branch tip 2. Node 2 splits to branch tips 1 and 3.
- Hypothesis 3. Root leads to node 1. Node 1 splits to node 2 and branch tip 1. Node 2 splits to branch tips 3 and 2.

Part 4. The three tree diagrams are updated to include the 1 C base change, noted in parentheses located at the following points.

- Hypothesis 1, node 2.
- Hypothesis 2, branch 1, branch 2.
- Hypothesis 3, branch 2, branch 1.

Part 5. Base change events are noted on each of the three tree diagrams as follows.

- Hypothesis 1. Node 2, 2 T. Branch 3, 3 A and 4 C. Branch 1, 3 A. Branch 2, 4 C.
- Hypothesis 2. Node 2, 3 A. Branch 2, 2 T and 4 C. Branch 1, 2 T. Branch 3, 4 C.
- Hypothesis 3. Node 2, 4 C. Branch 1, 2 T and 3 A. Branch 3, 3 A. Branch 2, 2 T.

Part 6. Base change events are totaled for each of the three trees with hash marks, indicating that Tree 1 is most parsimonious (requires the fewest number of base change events). Events are marked as follows.

- Hypothesis 1. Node 2, 2 events. Branch 3, 2 events. Branch 1, 1 event. Branch 2, 1 even. 6 total events.
- Hypothesis 2. Node 2, 1 event. Branch 2, 3 events. Branch 1, 2 events. Branch 3, 1 event. 7 total events
- Hypothesis 3. Node 2, 1 event. Branch 1, 3 events. Branch 3, 1 event. Branch 2, 2 events. 7 total events.

Part 3. A table is as follows.

blank	Site 1	Site 2	Site 3	Site 4
Species I	C	T	A	T
Species II	C	T	T	C
Species III	A	G	A	C
Ancestral sequence	A	G	T	T

- - STEP 1: First, draw the three possible trees for the species. (Although only three trees are possible when ordering three species, the number of possible trees increases rapidly with the number of species: There are 15 trees for four species and 34,459,425 trees for 10 species.)
- - STEP 2: Tabulate the molecular data for the species. In this simplified example, the data represent a DNA sequence consisting of just four nucleotide bases. Data from several outgroup species (not shown) were used to infer the ancestral DNA sequence.
- - STEP 3: Now focus on site 1 in the DNA sequence. In the tree on the left, a single base-change event, represented by the purple hatch mark on the branch leading to species I and II (and labeled 1/C, indicating a change at site 1 to nucleotide C), is sufficient to account for the site 1 data. In the other two trees, two base-change events are necessary.
- - STEP 4: Continuing the comparison of bases at sites 2, 3, and 4 reveals that each of the three trees requires a total of five additional base-change events (purple hatch marks).

Results

To identify the most parsimonious tree, we total all of the base-change events noted in steps 3 and 4. We conclude that the first tree is the most parsimonious of the three possible phylogenies. (In a real example, many more sites would be analyzed. Hence, the trees would often differ by more than one base-change event.)

Phylogenetic Trees as Hypotheses

This is a good place to reiterate that any phylogenetic tree represents a hypothesis about how the organisms in the tree are related to one another. The best hypothesis is the one that best fits all the available data. A phylogenetic hypothesis may be modified when new evidence compels systematists to revise their trees. Indeed, while many older phylogenetic hypotheses have been supported by new morphological and molecular data, others have been changed or rejected.

Thinking of phylogenies as hypotheses also allows us to use them in a powerful way: We can make and test predictions based on the assumption that a particular phylogeny—our hypothesis—is correct. For example, in an approach known as *phylogenetic bracketing*, we can predict (by parsimony) that features shared by two groups of closely related organisms are present in their common ancestor and all of its descendants unless independent data indicate otherwise. (Note that "prediction" can refer to unknown past events as well as to evolutionary changes yet to occur.)

This approach has been used to make novel predictions about dinosaurs. For example, there is evidence that birds descended from the theropods, a group of bipedal dinosaurs. As seen in **Figure 26.16**, the closest living relatives of birds are crocodiles. Birds and crocodiles share numerous features: They have four-chambered hearts, they "sing" to defend territories and attract mates (although a crocodile's "song" is more like a bellow), and they build nests. Both birds and crocodiles also care for their eggs by warming them. Birds warm their eggs by sitting on them, whereas crocodiles cover their eggs with their neck or with sand or vegetation. Reasoning that any feature shared by birds and crocodiles is likely to have been present in their common ancestor (denoted by the blue dot in Figure 26.16) and *all* of its descendants, biologists predicted that dinosaurs had four-chambered hearts, sang, built nests, and cared for their eggs.

Figure 26.16

A phylogenetic tree of birds and their close relatives.

A phylogenetic tree.

The tree is as follows. Root leads to node 1. Node 1 splits to branch tip lizards and snakes and node 2. Node 2 represents common ancestor of crocodilians, dinosaurs, and birds. It splits to branch tip crocodilians and

node 3. Node 3 splits to node 4 and branch tip other dinosaurs. Node 4 spits to branch tips birds and theropod dinosaurs other than birds.

(A dagger, †, indicates extinct lineages.)

VISUAL SKILLS *In this tree diagram, what is the sister taxon of the clade that includes dinosaurs and their most recent common ancestor? Explain.*

*For suggested answer, see*Appendix A.**

Internal organs, such as the heart, rarely fossilize, and it is, of course, difficult to test whether dinosaurs sang to defend territories and attract mates. However, fossil discoveries have supported the prediction that dinosaurs built nests and cared for their eggs. First, a fossil embryo of an *Oviraptor* dinosaur was found, still inside its egg. This egg was identical to those found in another fossil, one that showed an *Oviraptor* crouching over a group of eggs in a posture similar to how birds sit on their nests today **(Figure 26.17)**. Researchers suggested that the *Oviraptor* dinosaur preserved in this second fossil died while incubating or protecting its eggs. The broader conclusion that emerged from this work—that dinosaurs built nests and cared for their eggs—has since been strengthened by additional fossil discoveries that show that other species of dinosaurs built nests and sat on their eggs. Finally, by supporting predictions based on the phylogenetic hypothesis shown in Figure 26.16, fossil discoveries of nests and parental care in dinosaurs provide independent data that suggest that the hypothesis is correct.

Figure 26.17

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    Fossil support for a phylogenetic prediction: Dinosaurs
built nests and cared for their eggs.
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Fossil remains of Oviraptor and eggs. The orientation of the bones, which surround and cover the eggs, suggests that the dinosaur died while incubating or protecting its eggs.